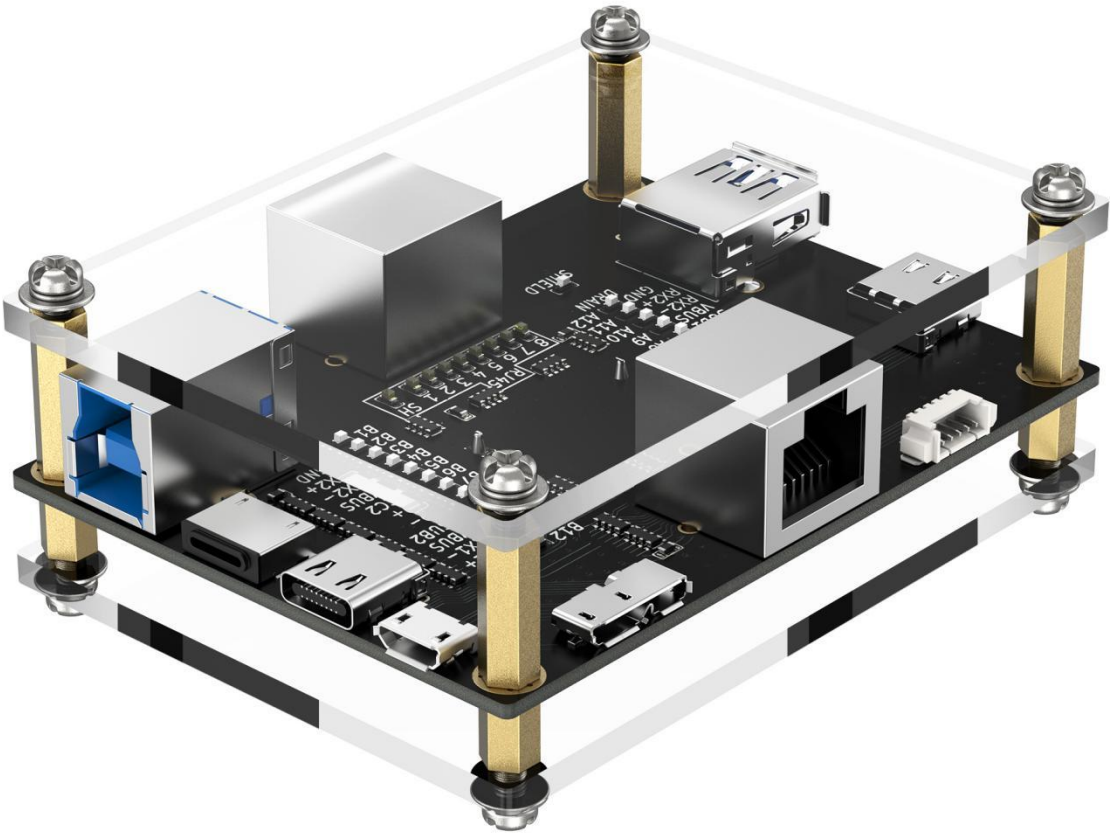
CableLinkCheck UserManual



1. General Description:

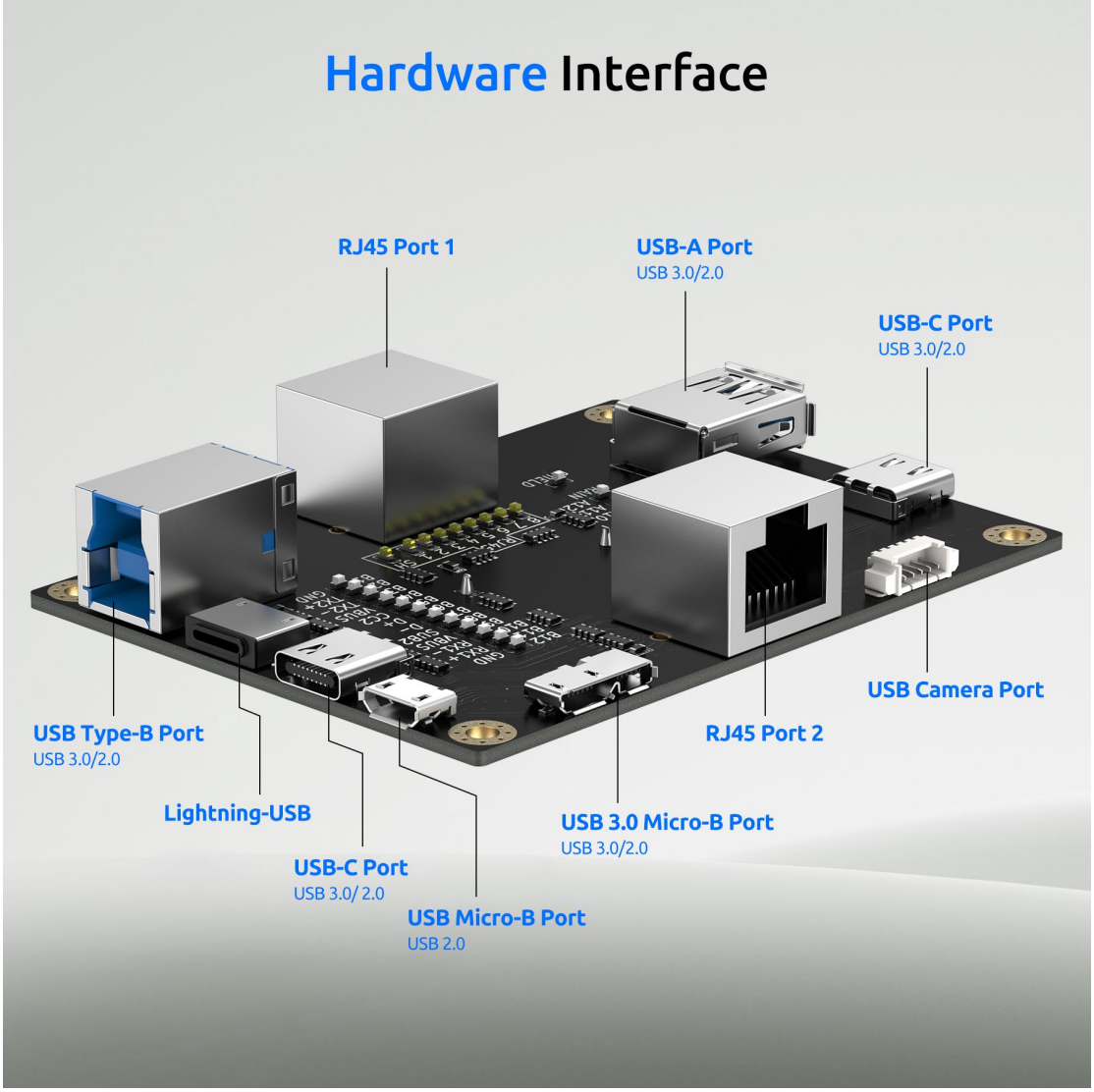
CableLinkCheck is a portable continuity tester with integrated USB-A, USB-C, Micro USB, and RJ45 ports. Onboard LEDs allow users to visually inspect cable connections. It is designed for DIY projects, electronic repair, cable quality checks, and network cable maintenance.

The tester supports basic continuity checks for compatible USB 2.0 / USB 3.0 cables and CAT5/CAT6 RJ45 Ethernet cables. All signal lines in USB 2.0/3.0 are fully broken out, each with a corresponding LED indicator. RJ45 indicators marked 1-8 and SH help verify continuity of the eight conductors and the shield. No software installation is required. Built-in Battery Power Supply.

Please note: Do not plug any powered USB connector into the board, and do not connect it to any port on a computer.

2. Hardware

2.1 Interface



2.1 Interface

PCB Marking	USB Standard	Signal Description
GND	USB 3.0/2.0	Ground connection
VBUS	USB 3.0/2.0	USB power supply (+5V)

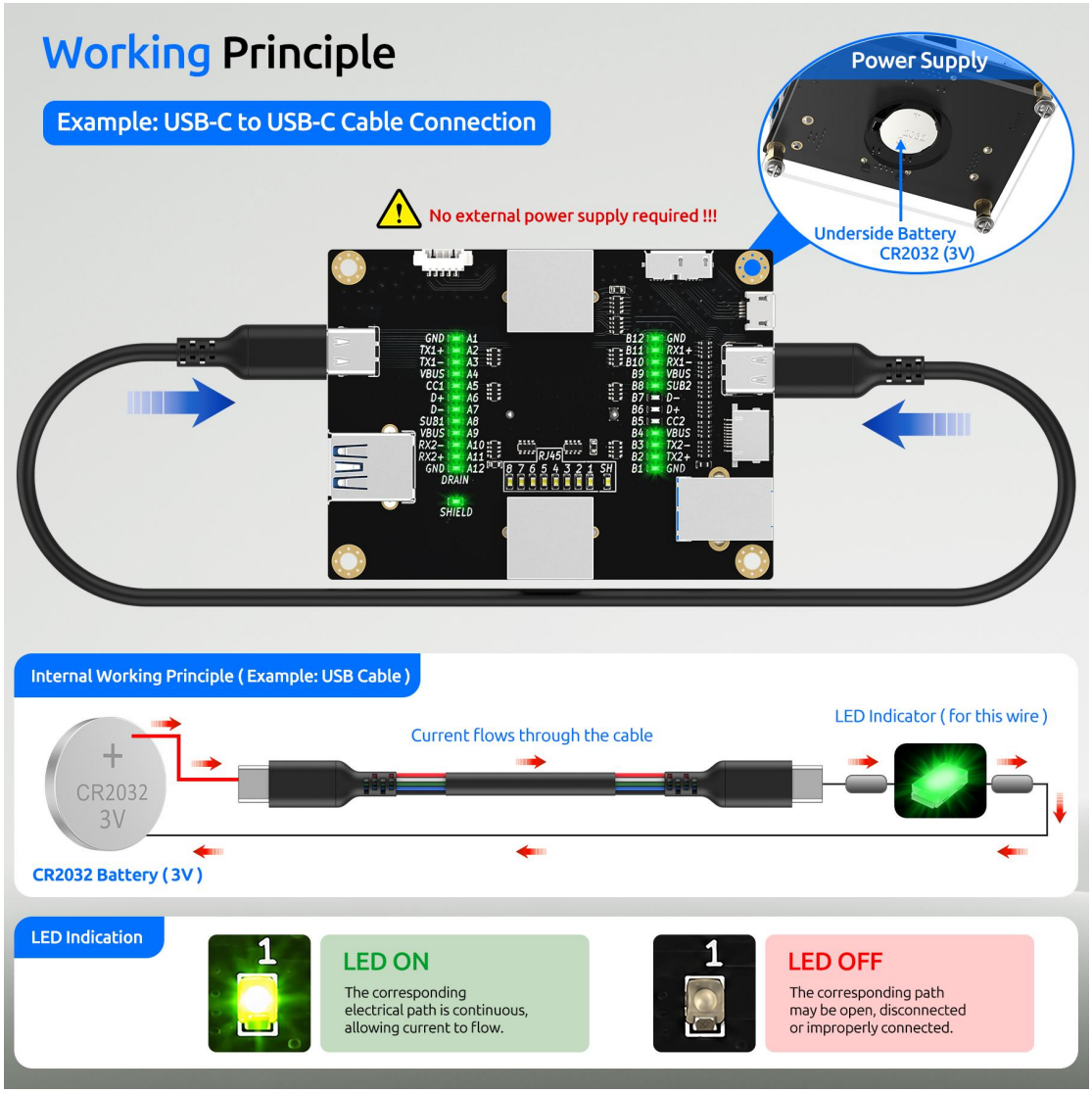
TX1+	USB 3.0	SuperSpeed transmit differential pair 1
TX1–	USB 3.0	SuperSpeed transmit differential pair 1
RX1+	USB 3.0	SuperSpeed receive differential pair 1 (+)
RX1–	USB 3.0	SuperSpeed receive differential pair 1 (–)
TX2+	USB 3.0	SuperSpeed transmit differential pair 2
TX2–	USB 3.0	SuperSpeed transmit differential pair 2
RX2+	USB 3.0	SuperSpeed receive differential pair 2 (+)
RX2–	USB 3.0	SuperSpeed receive differential pair 2 (–)
D+	USB 2.0	USB 2.0 differential data (+)
D–	USB 2.0	USB 2.0 differential data (–)
CC1	USB Type-C	Configuration Channel 1
CC2	USB Type-C	Configuration Channel 2
SBU1	USB Type-C	Sideband Use 1
SBU2	USB Type-C	Sideband Use 2
SHIELD	USB 3.0/2.0	Cable shield

PCB Marking	RJ45 Pin	Signal Description (T568B)
1	Pin 1	White/Orange – Pair 2
2	Pin 2	Orange – Pair 2
3	Pin 3	White/Green – Pair 3
4	Pin 4	Blue – Pair 1
5	Pin 5	White/Blue – Pair 1
6	Pin 6	Green – Pair 3
7	Pin 7	White/Brown – Pair 4
8	Pin 8	Brown – Pair 4
SH	Shield	RJ45 connector shield / cable shielding

3. Working Principle -Battery-Powered Continuity Testing

The CableLinkCheck Cable Tester operates on a simple battery-powered continuity testing principle. A built-in battery located on the back of the circuit board provides the electrical power required for testing. When both ends of a USB cable or RJ45 Ethernet cable are connected to the corresponding test ports, the internal battery supplies a small electrical current through the cable's conductive wires. If a wire is properly connected and continuous, it completes an electrical circuit, allowing current to flow and illuminating the corresponding LED indicator.Each LED indicator represents a specific wire or signal path, enabling users to visually check the continuity of individual connections.

LED Indication:
LED ON: The corresponding electrical path is continuous, allowing current to flow.
LED OFF: The corresponding path may be open, disconnected, or improperly connected.

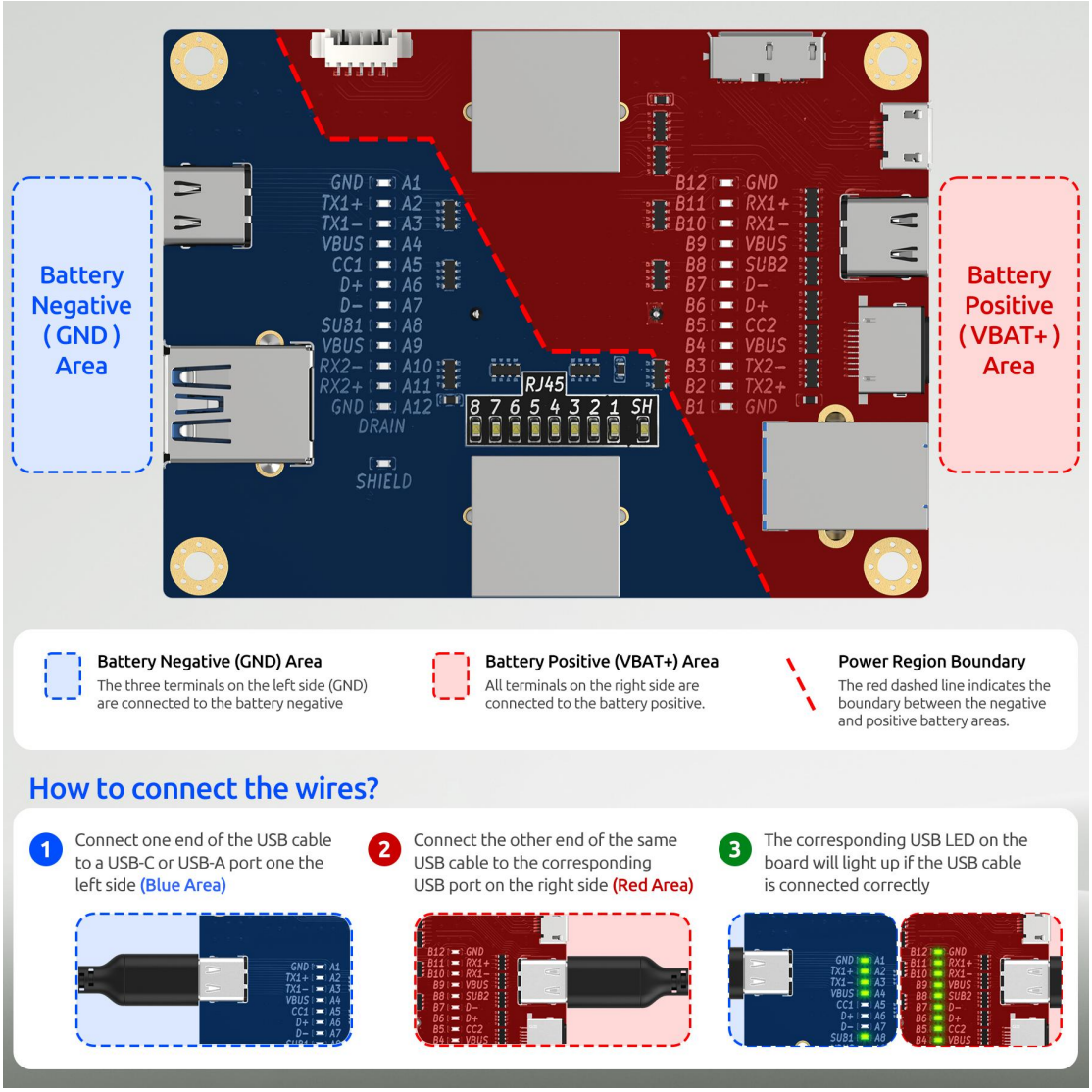


The tester requires no external power supply, computer, software, or additional testing equipment. Simply connect both ends of the cable to the appropriate ports and observe the LED indicators.

Note: This product is designed for basic cable continuity and wiring checks. It does not measure data transmission speed, signal quality, or USB Power Delivery performance.

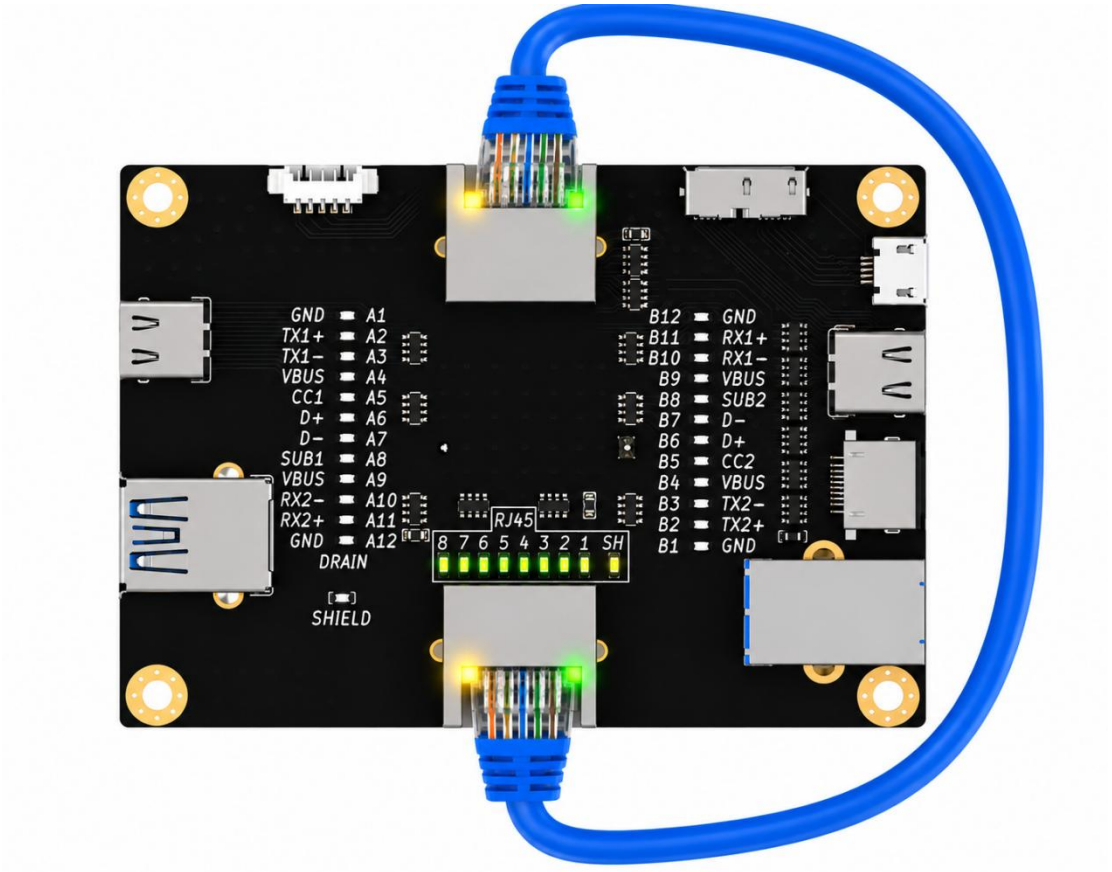
4. Testing USB Cable

Insert one end of the USB cable to be tested into either the USB-C or USB-A port in the blue area on the left side of the board. Then plug the other end of the USB cable into the corresponding USB port in the red area. The corresponding LED will light up.



5. Testing RJ45 Ethernet Cables

Before testing, disconnect the routers, switches, computers, and PoE power supply equipment connected to both ends of the cable. Insert the two cable ends into the upper and lower RJ45 ports on the tester. Observe the 1 – 8 and SH indicator areas. For a straight-through cable wired to the same standard at both ends, the indicators should normally map 1→1 through 8→8; for a crossover cable, judge according to the actual wiring sequence. SH is used to help check shield continuity; for UTP unshielded cable, SH not lighting is usually normal. Eight-core continuity does not mean the cable meets all CAT5/CAT6 performance requirements; professional certification still requires tests for attenuation, crosstalk, return loss, and other parameters.



4. User Manual Version Descriptions

Version	Description	Date	E-mail
V1.0		2026.05.10	support@inno-maker.com sales@inno-maker.com

If you have any suggestions, ideas, codes and tools please feel free to email to me. I will update the user manual and record your name and E-mail in list. Look forward to your letter and kindly share.